Build the Executive OS decision layer for marketplace and partner performance metrics — take rate, GMV, churn, and partner ROI on a live dashboard

Computer
Turn the Executive OS marketplace outline into a strategic KPI scorecard for leadership review — partner health, tier migration, and unit economics
Computer
Design the marketplace metrics layout
Define the partner performance KPIs
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Built two artifacts: a working Marketplace & Partner Decision Layer dashboard and its standalone JavaScript source. The dashboard includes GMV, realized take rate, contribution, partner exit rate, partner ROI, tier movements, service performance, trust status, and a leadership decision brief.

It is **live-ready**, not live-connected: it starts with clearly labeled synthetic data, accepts JSON snapshots, and includes a same-origin adapter that can poll every 60 seconds once hosted inside Semester and connected to a verified, authorized backend. No API, repository change, deployment, payment, tier promotion, or authenticated approval was created.

1. Marketplace metrics layout

The dashboard follows a decision-oriented layout rather than a raw reporting table.

A. Evidence and scope

At the top:

- Reporting period and currency.
- Synthetic versus connected-snapshot indicator.
- Source/freshness description.
- Complete versus incomplete snapshot status.
- Partner-tier filter.

- Partner search.
- Sorting by contribution, net GMV, ROI, or name.
- Snapshot import and adapter controls.

All headline cards recalculate for the selected partners. The screen does not mix currencies.

Production should additionally expose metric version, warehouse watermark, last complete load, source lineage, and approved reporting scope. Semester's architecture requires canonical models, shared permissions, audit, and workflow controls rather than disconnected dashboards. [1]

B. Leadership headline cards

Card	Decision it supports
Gross GMV	Is eligible marketplace transaction volume growing?
Net GMV	How much volume remains after aligned refunds?
Net transaction fees	What transaction-based fees does Semester retain?
Realized take rate	How effectively is that volume monetized?
Contribution	Does transaction activity cover its variable costs?
Starting-cohort exit rate	Are existing partners leaving?

GMV measures marketplace transaction volume, not the revenue retained by the platform. High GMV does not establish profitability. [2] [3]

C. Partner scorecard

Each row shows:

```
Partner
Previous tier → current tier
Net GMV
Realized take rate
Contribution
Program ROI and estimate basis
SLA attainment
Trust-review status
```

Selecting a partner opens the decision brief.

D. Leadership decision brief

The brief contains:

- Evidence summary.
- Suggested review category.
- Alternatives and tradeoffs.

- Accountable owner.
- Review date.
- Exportable draft.

The rules are illustrative review routing—not approved institutional policy:

- Incomplete snapshot → block recommendations.
- Trust issue → review before expansion.
- Exited partner → examine exit and portability obligations.
- Negative contribution → review economics.
- Negative estimated program ROI → review investment.
- Otherwise → prepare growth review, without automatically promoting the partner.

2. Partner performance KPI definitions

Marketplace terminology varies between sources—for example, whether cancellations are included in a reported GMV figure. Semester should therefore publish a versioned metric contract rather than rely on labels alone. [4] [5]

The following are proposed Semester definitions, implemented in the demonstration.

Transaction volume and monetization

KPI	Proposed formula	Important boundary
Eligible gross GMV	Sum of eligible fulfilled transaction values after seller discounts, before refunds	Exclude taxes, shipping, unfulfilled/cancelled transactions under this contract
Net GMV	Eligible gross GMV − refunds for that same transaction cohort	Do not subtract unrelated prior-period refunds without labeling the method
Net transaction fees	Transaction fees − fee reversals	Exclude subscriptions and unrelated revenue
Realized take rate	Net transaction fees ÷ net GMV	Ratio of totals, not the average of partner percentages
Refund rate	Aligned refund value ÷ eligible gross GMV	Distinguish refund dollars from refunded-order count
Payment-settled volume	Eligible transactions confirmed settled	Separate from booked or fulfilled volume

Take rate expresses the portion of transaction value retained as fees or revenue. The denominator and fee treatment must remain explicit. [6] [7]

The dashboard implements the first four metrics. Refund rate and payment-settled volume are recommended production extensions.

Unit economics

$$
\text{Transaction contribution}
$$

Net transaction fees

$$
\text{Variable transaction costs}
$$

Variable costs should follow an approved allocation policy—for example, processing, transaction-specific support, delivery infrastructure, fraud losses, or attributable AI usage.

The demo keeps partner-program investment separate from transaction contribution to avoid subtracting it twice.

Production extensions:

KPI	Formula or basis
Contribution per fulfilled order	Transaction contribution ÷ fulfilled orders
Contribution margin	Transaction contribution ÷ net transaction fees
Fully loaded partner contribution	Transaction contribution − approved partner-program and allocated operating costs
Partner acquisition cost	Attributable acquisition/onboarding spend ÷ activated partners
Payback	Time until cumulative attributable contribution covers acquisition investment
Concentration	Share of net GMV or contribution produced by the largest partners

Do not present an LTV estimate until retention, margins, observation periods, and assumptions are sufficiently supported.

Partner churn

The demonstration implements an explicit exit measure:

$$
\text{Partner exit rate}
$$

$$
\frac{\text{Starting-active partners that exited}}{\text{Starting-active partners}}
$$

Newly activated partners are excluded from that denominator.

Keep these different measures separate:

- Contractual exit: relationship formally ended.

- Transaction inactivity: no qualifying activity within a defined window.
- Installation churn: application uninstalled or authorization revoked.
- Revenue contraction: retained partner generates less revenue.
- Tier downgrade: approved tier changed, but the partner may still be active.

An inactive partner is not necessarily churned, particularly in an academic marketplace with term-related seasonality.

Partner ROI

The implemented metric is Semester's estimated return on a partner-program investment—not the partner organization's own return.

[
Program ROI

\frac{
Incremental contribution before program cost

Program cost
}{
Program cost
}
]

ROI compares a gain or net return with investment cost; using incremental profit/contribution is different from reporting revenue divided by spend. [8]

Required fields:

- Investment scope and period.
- Included costs.
- Baseline or comparison method.
- Incremental contribution estimate.
- Attribution limitations.
- Confidence or review status.
- Reviewer.

The screen displays an ROI-basis label. If the incremental estimate is unavailable—or investment is zero—it shows "Unavailable," not a fabricated percentage.

A partner-facing ROI metric requires a different numerator: the partner's attributable margin or verified savings, less its investment. Do not label Semester's program ROI as the partner's ROI.

3. Partner health and tier migration

Health should be a profile, not one opaque number

Dimension	Example evidence	Leadership response
Economics	Contribution, refunds, costs, ROI	Pricing, investment, or remediation review
Service	SLA attainment, incidents, unresolved work	Capacity or corrective-action plan
Adoption	Activation and completed workflows	Enablement review
Trust	Scope review, obligations, security findings	Restrict expansion pending review
Relationship	Sponsor engagement, unresolved commitments	Account-owner intervention
Data confidence	Freshness, completeness, reconciliation	Defer unsupported conclusions

The built screen includes economics, SLA attainment, trust status, and data completeness. Adoption and relationship evidence should be added through authorized source contracts.

Tier movements

The demonstration displays previous and current tiers from the input snapshot. It does not infer or execute promotion.

A production tier workflow should be:

```
Eligibility evidence
  → policy evaluation
  → impact preview
  → required review
  → approved effective-dated change
  → entitlement/configuration update
  → confirmation
  → periodic reassessment
```

Proposed movement categories:

- Promotion.
- Downgrade.
- Unchanged.
- New activation.
- Exit.
- Re-entry.

Trust or contractual issues should not disappear inside a weighted growth score. Expansion should remain a human-authorized, policy-controlled workflow. Semester's specifications similarly separate analysis/drafting from financial approvals and require authorization beyond the route boundary.[1]

4. Code delivered

The standalone JavaScript source contains:

- Snapshot validation.
- Duplicate-partner detection.
- Finite-number and denominator handling.
- Currency formatting.
- Filtered aggregate calculations.
- Partner sorting and scorecards.
- Observed tier movements.
- Decision-review routing.
- JSON import.
- Same-origin polling.
- Failed-refresh preservation of the prior snapshot.
- Draft review export.

The central metric function is:

```
function metrics(partner) {
  const netGMV = partner.gmv - partner.refunds;
  const netFees = partner.fees - partner.feeReversals;
  const contribution = netFees - partner.variableCosts;

  return {
    netGMV,
    netFees,
    contribution,

    takeRate:
      netGMV > 0 ? netFees / netGMV : null,

    contributionMargin:
      netFees > 0 ? contribution / netFees : null,

    programROI:
      partner.incrementalContributionBeforeProgram === null ||
      partner.programCost <= 0
        ? null
        : (
            partner.incrementalContributionBeforeProgram -
            partner.programCost
          ) / partner.programCost,

    slaAttainment:
      partner.slaTotal > 0
        ? partner.slaMet / partner.slaTotal
        : null
```

```
  };
}
```

The delivered source uses equivalent calculations, with shared helpers and rendering logic.

Snapshot contract

The screen expects:

```
schemaVersion
mode: synthetic | live
currency
period
freshness
complete
partners[]
```

Each partner contains:

```
id, name
tier, previousTier
startActive, exited
gmv, refunds
fees, feeReversals
variableCosts
programCost
incrementalContributionBeforeProgram
roiBasis
orders
slaMet, slaTotal
trust
```

This is a proposed UI adapter contract, not a new canonical database schema.

5. Connecting it to live Semester data

To make the dashboard operational:

1. Reconcile the contract with existing marketplace, commercial, partner, and ledger records.
2. Build an authenticated, read-only aggregate adapter.
3. Resolve permitted company/institution scope server-side.
4. Enforce authorization before aggregating or returning data.
5. Return one currency and a defined reporting cohort.
6. Include completeness, freshness, metric version, and source evidence.
7. Host the screen inside Semester's existing app shell.
8. Enter the verified same-origin adapter path.
9. Enable polling and validate refresh/failure behavior.

10. Keep tier changes and financial actions in the existing controlled-command workflow.

The adapter must not trust a browser-provided tenant identifier as authorization. The dashboard also cannot independently verify whether a response labeled `live` is authoritative.

Acceptance gates

- Aggregates cannot leak unauthorized partners.
- Take rate uses a ratio of totals.
- Exited starting-cohort partners remain in churn calculations.
- Zero denominators display “Unavailable.”
- Refunds and fees use aligned cohorts.
- Incomplete data blocks recommendation generation.
- Refresh failure preserves—but clearly identifies—the prior snapshot.
- Tier history comes from approved effective-dated records.
- ROI estimates expose their basis and limitations.
- Exported briefs remain drafts until authenticated review.
- Financial calculations reconcile to approved source records.
- Keyboard, loading, empty, error, and narrow-screen behavior receive testing.

The dashboard and source files were generated and verified as non-empty. They have not been browser-tested, integrated, deployed, or connected to a live datasource in this conversation.

⁂

1. The Main Semester design system (2) copy/handoff/specifications/role-system-execution-prompt.pdf
2. https://kruzeconsulting.com/blog/gmv-gross-merchandise-value/
3. https://www.crv.com/content/gmv-meaning
4. https://recurly.com/glossary/gross-merchandise-value-gmv/
5. https://www.metabase.com/metrics/gmv
6. https://www.opendue.com/glossary/take-rate
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9. https://bluecopa.com/blog/guide-to-understanding-gross-merchandise-value-gmv-for-online-marketplace
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